

Collaborative World Models, Semantic Evidence Preservation, and Test-Time Adaptation

Statement of Purpose and Postdoctoral Research Plan

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1. Research Vision: From Isolated Encoders to Collaborative World Models

A fundamental frontier in artificial intelligence is building visual systems that do not merely categorize static snapshots, but understand how environments evolve, maintain persistent spatial and temporal memories, and communicate across perspectives to solve embodied problems. While foundation models encode immense visual and semantic knowledge from web-scale data, two core bottlenecks persist:

- 1. Isolated vs. Collaborative Modeling:** Current world models are predominantly trained as monolithic, single-stream predictors. In contrast, real-world embodied perception requires collaborative intelligence: decentralized agents operating from disparate physical viewpoints, maintaining specialized memory banks, and reconciling localized observations into a coherent spatio-temporal belief state.
- 2. Premature Collapse during Test-Time Adaptation:** When foundation representations adapt to novel or shifting environments, standard architectures commit prematurely to hard, deterministic predictions (argmax), permanently discarding secondary hypotheses and preventing downstream consensus.

My doctoral research has investigated the theoretical and algorithmic foundations of this challenge: formulating non-parametric inference-time coordination ($\Delta\theta = 0$), proving the benefits of semantic evidence preservation, and deriving label-free process rewards from multi-view physical consistency. At the Computer Vision Group (CVG) at the University of Bern, in collaboration with the University of Oxford, I propose to bridge these insights to develop **Collaborative World Models** that maintain distinct, calibrated memory streams, communicate via geometric-semantic consensus, and achieve rapid test-time adaptation in complex physical environments.

2. Doctoral Research Highlights and Core Discoveries

My doctoral dissertation, *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations* (supervised by Prof. Ebenezer Owusu), provides three methodological pillars directly relevant to the University of Bern research program:

2.1 Reliability-Calibrated Semantic Arbitration (RCASA) and Distributional Coordination

In my thesis work (submitted to IEEE TPAMI), I formulated a probabilistic framework for coordinating heterogeneous, independently pretrained foundation models without parameter updates. By formally separating source confidence from contextual correctness through Platt scaling and Bernoulli reliability weighting, the framework establishes that multi-model interaction can be executed non-parametrically at inference time. This provides the mathematical machinery needed for collaborative visual agents to arbitrate trust and aggregate observations dynamically without continual fine-tuning.

2.2 The Physics of Semantic Preservation: Overcoming Premature Collapse

A central finding of my thesis is that premature decision collapse—discarding sub-maximal semantic hypotheses at the output of individual visual encoders—severely constrains downstream arbitration. In controlled experiments on the 312 scenes (~ 50 M points) of ScanNet200 and ScanNet++, I demonstrated that retaining compact Top- k candidate distributions allows downstream consensus to recover previously overlooked classes, boosting segmentation harmonic mean from 28.47 to 34.87 (+6.40 points). For world models predicting environmental transitions, preserving uncertainty and

alternative hypotheses over time is crucial for preventing compounding errors in generative rollouts.

2.3 Multi-View Geometric Lifting and Label-Free Process Supervision

In ACCV 2026 (Paper #482) and recent reinforcement learning work submitted to SACAIR 2026, I constructed 2D-to-3D geometric lifting pipelines connecting camera poses (K, T) with multi-view point-cloud consensus. Furthermore, we showed that physical agreement across independent view-points provides a verifiable, label-free process reward for training multimodal LLMs (Qwen2.5-VL via GRPO/VERL on vLLM), eliminating area-bias gaming ($\rho > 0.72 \rightarrow 0.334$) without human-labeled reward models.

3. Proposed Research Agenda at CVG Bern (with Univ. of Oxford)

At the University of Bern, I aim to expand these foundations within the **Collaborative World Models** and **Rapid Adaptation** initiatives:

Theme A: Decentralized Spatio-Temporal Consensus for Multi-Agent World Models

Monolithic world models struggle with partial occlusions, varying sensor resolutions, and dynamic multi-agent settings. I propose to develop a decentralized consensus protocol where independent visual agents maintain local 3D/video memory representations (e.g., dynamic Gaussian Splats or continuous feature fields) and exchange compact, calibrated semantic-geometric hypotheses. Using probabilistic consensus rules grounded in Hoeffding bounds, agents will negotiate a shared spatial-temporal representation of dynamic environments without requiring centralized retraining.

Theme B: Test-Time Adaptation via Distribution-Preserving Memory Rollouts

Current generative video and world models often suffer from drift and hallucination during multi-step rollout planning. Drawing on my discovery of premature semantic collapse, I propose to formulate *distribution-preserving latent rollouts*. Rather than sampling single trajectories greedily, the model will track multi-hypothesis distributions through time, using test-time adaptation (e.g., entropy regularization and self-supervised consistency losses) to prune improbable futures as new visual evidence is observed.

Theme C: Physical Geometry as Self-Supervised World-Model Objectives

To minimize reliance on massive labeled datasets, I will extend my label-free process-reward framework to video world models. By enforcing multi-view epipolar consistency, photometric reprojection invariance, and 3D geometric continuity across predicted video frames, we can train embodied world models using self-supervised spatial priors, enabling rapid transfer to robotics and navigation benchmarks.

4. Institutional Synergies: University of Bern and University of Oxford

The Computer Vision Group at the University of Bern, led by Prof. Paolo Favaro, has established world-class leadership in self-supervised representation learning, inverse problems, and deep generative architectures. The collaboration with Prof. Andrea Vedaldi at the University of Oxford creates an unmatched ecosystem for foundational research in visual representations.

My technical strengths in PyTorch, 3D point-cloud and geometric transformations, CUDA-accelerated inference, and empirical statistical methodology align naturally with the experimental demands of these projects. I am deeply motivated to join CVG Bern, collaborate across both institutions, and contribute to pushing the boundaries of collaborative, adaptive, and physically grounded world models.